

Reflex Computation: Semantic Execution Blocks and Interface-Knowledge Placement

Working Draft

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Abstract

Language models can describe arithmetic procedures while still making exact execution errors. Conventional tool-use systems address this weakness by requiring the model to decide to call a calculator and serialize an invocation. We study bounded local calculator interfaces that act during generation. The original strict watcher recognizes an arithmetic suffix before an equals sign; the next iteration adds deterministic LaTeX/Unicode surface normalization and a fenced calculator block that executes only when its closing delimiter arrives. All assisted conditions share the same typed postfix micro-IR, exact-rational engine, resource bounds, and textual reinjection. After baseline-only adaptive selection, an XPU comparison used 16 arithmetic examples, 12 controls, and seven conditions. Exact-match accuracy was 43.8% for LLM-only, 62.5% for explicit tools, 75.0% for normalized reflex, and 12.5% for the original zero-shot block; normalized reflex gained five paired examples and lost none ($p = 0.0625$, unadjusted exact McNemar). We then developed block prompts on a separate split and froze a negative-first, positive-last two-shot protocol. On the same diagnostic examples, frozen blocks reached 100% valid execution and arithmetic accuracy with Qwen3-0.6B, but falsely blocked 2 of 12 controls. Qwen3-1.7B reached 93.8% execution and accuracy with zero false blocks, gaining nine and losing one pair against LLM-only ($p = 0.0215$). A Qwen3-4B six-item XPU smoke reached 100% execution with no false block, but is not a held-out estimate. Protocol-only rank-8 LoRA adapters were then trained for Qwen3-0.6B, SmolLM2-1.7B, and the approved official Gemma3-1B checkpoint. With a minimal instruction, all three adapters reached 100% block execution with zero false blocks; answers were 100%, 100%, and 93.8%, respectively. Gemma reached 100% answers when ICL was added, whereas its base ICL and normalized-runtime conditions executed only 31.2% and 25.0%. These developmental results favor a hybrid: store stable semantic selection and serialization in small adapter weights, keep exact execution and enforced result use in the runtime, and reserve context for cold-start or model-specific residual behavior. The sample remains too small for a general scaling or family claim. A frozen 120-item GSM8K transfer stress test then reversed the synthetic result: LLM-only and an unexecuted block prompt both reached 10.8% accuracy, generic compute 2.5%, automatic runtime 3.3%, and LoRA block 0.8%, while an oracle bounded-calculator ledger reached 75.8%. Automatic Rescue Rate was 0/6. Thus interface syntax learned on synthetic data did not supply the natural-language decomposition needed for public transfer.

1 Status and Scope

This document reports an implemented next-iteration protocol, the preliminary pilot that motivated it, adaptive calibration, and one held-out comparison. The repository contains a dependency-free calculator/runtime path, deterministic dataset generation, JSONL traces and summaries, a scripted plumbing check, and an optional token-by-token Hugging Face adapter. Scripted checks are marked `empirical: false`; their expected perfect calculator behavior must not be cited as

model evidence. The XPU pilot reported below is empirical but too small for model-family, scaling, or replicated performance claims. The held-out run is empirical and protocol-frozen, but it also has one checkpoint and one greedy seed; its condition ordering is evidence for replication, not a final architecture-level estimate. The public GSM8K slice is likewise a single-checkpoint developmental transfer test, not a benchmark-wide estimate.

Paper 1 asks one question:

When a language model naturally exposes a completed arithmetic expression during generation, does automatic bounded evaluation improve reliability or cost relative to model-only generation and explicit calculator calls?

The scope remains calculator only and deterministic syntax interfaces. The experiment does not test semantic recognition of prose arithmetic, learned routing, persistent symbolic memory, transactional state, Progressive Retrieval Attention (PRA), or native key/value access. Those mechanisms would make a positive result harder to attribute and a negative result harder to diagnose.

2 Hypotheses

H1: exact reliability. On arithmetic examples for which the strict event is correctly detected and normalized, reflex assistance increases exact-match accuracy relative to the same model without assistance.

H2: interface efficiency. Reflex or fenced-block assistance reduces model-authored invocation overhead relative to an explicit calculator condition while remaining competitive in answer quality and wall time.

H3: difficulty scaling. As operator count and operand width increase, reflex accuracy degrades more slowly than model-only and matched-prompt conditions. The relevant evidence is the condition-by-difficulty interaction or fitted scaling slope, not one easy benchmark cell.

H4: model-size leverage. Smaller models receive at least as much marginal accuracy benefit from correct reflex assistance as larger matched-family models. This is tested as a model-size-by-condition interaction, not inferred from separate unrelated checkpoints.

These hypotheses are conditional on detection and use. Oracle detection estimates interface headroom; engine correctness alone does not demonstrate that the model will expose the syntax or use the inserted result.

3 Mechanism

3.1 Strict generation-time event

The reflex recognizes an integer arithmetic suffix completed by one equals sign, for example

The total is $37 * 48 =$

and inserts the exact result before later generated text:

The total is $37 * 48 = 1776$.

Table 1: Default calculator safety and complexity bounds.

Bound	Default
Expression characters	256
AST nodes / depth	64 / 16
Digits per integer literal	64
Absolute integer exponent	12
Numerator or denominator size	512 bits
Stack items	64
Engine time budget	25 ms
Recognizer suffix buffer	512 characters

The event is ordinary written arithmetic, not a bespoke API token. Detection is incremental at character granularity even when a tokenizer supplies multi-character fragments. Thus, if an equals sign occurs in the middle of a fragment, the result is inserted before the remainder of that fragment is processed.

The lexical detector accepts digits, whitespace, parentheses, and `+`, `-`, `*`, `/`, `//`, `%`, and `**`. It requires an operator-like symbol, balanced parentheses, and a digit or closing parenthesis before the equals sign. It suppresses double-quoted candidates and rejects candidates adjacent to letters, underscores, or decimal points. Lexical detection remains separate from normalization: a candidate such as `+ 2` is logged and then rejected because it has no binary operation.

3.2 Typed micro-IR

The normalizer parses the candidate in expression mode and accepts only integer constants, binary operations, unary plus, and unary minus. Calls, names, attributes, indexing, containers, comparisons, booleans, floats, and all other syntax are rejected. The parsed tree is converted to a canonical postfix instruction sequence such as

PUSH 7; PUSH 5; ADD; PUSH 3; MUL.

The Python parser is used only to construct and inspect syntax. The source is never compiled or executed. A separate stack interpreter executes the micro-IR using exact rational numbers.

3.3 Resource policy

Every event is bounded before or during evaluation. The default policy is shown in Table 1. Bounds are configuration recorded in each request, not hidden assumptions.

Division by zero, malformed IR, timeout, and resource exhaustion are typed failures. Failed results are traced but not inserted. Exponent and intermediate bit limits are checked before a large value can propagate through subsequent instructions.

3.4 Observable interrupt pipeline

Each intervention follows

DETECT $\rightarrow$ NORMALIZE $\rightarrow$ ROUTE $\rightarrow$ EXECUTE $\rightarrow$ STATE UPDATE $\rightarrow$ REINJECT.

Every transition receives a run ID, sequence number, timestamp, status, candidate/request IDs, duration where meaningful, and structured details. The append-only micro-state stores the typed

Table 2: Minimum experimental conditions.

Condition	Model instruction and capability	Purpose
LLM only	Return the exact answer; no calculator	Unassisted checkpoint baseline
Matched prompt	Work carefully and check operations; no calculator	Stronger prompting without external capability
Explicit tool	Model writes <code><tool:calculator>EXPR</tool></code> and receives the same engine result	Conventional model-authored invocation baseline
Reflex	Model is asked to write the expression followed by one equals sign; runtime detects any valid strict suffix	Proposed automatic condition
Normalized reflex	Reflex syntax plus deterministic LaTeX, Unicode, whitespace, and bracket normalization	Surface-robust automatic condition
Calculator block	Model writes one fenced calculator block; execution waits for the closing fence	Explicit boundary without an API schema
Oracle	Same generation instruction as reflex, but the detector accepts only the gold canonical expression	Detection/selection headroom control

request, result, creation time, detector, and engine provenance. This is enough to separate detector, parser, engine, and reinjection failures without importing the later transactional-state design.

4 Conditions

All conditions use the same arithmetic examples, checkpoint revision, decoding budget, and calculator implementation when a calculator is available.

The explicit format is deliberately unambiguous and is not counted as a reflex. Its parser and the reflex parser converge on the same arithmetic micro-IR and engine, isolating invocation style from calculator capability. A stronger native function-calling baseline should be added when the selected model/runtime exposes a stable function-calling interface; the textual baseline is the portable minimum.

5 Benchmark

5.1 Factorial arithmetic generators

The legacy deterministic generator crosses operator count with operand digit width. The separate `hard_arithmetic_v1` generator crosses at least four operator-count bands, four digit-width bands, and three structures: left chains, nested trees, and multiplication trees. It includes addition, subtraction, multiplication, safe exact-rational division, mixed operations, and long intermediates. Each cell uses identity-derived random seeds, and gold answers are produced by a separate exact-rational reference evaluator rather than the calculator implementation under test. The full hard grid contains

$$4 \text{ operator counts} \times 4 \text{ digit widths} \times 3 \text{ structures} = 48 \text{ cells}$$

before replicate and seed factors. Every generated item is accepted only after the independent oracle and bounded calculator agree.

This synthetic benchmark isolates scaling and component correctness; it does not establish ecological prevalence. A later extension may add held-out natural model traces, but only after freezing annotation rules for whether a generated prefix contains a valid strict event.

5.2 Non-trigger and adversarial controls

Controls include prose containing numbers, ranges, parentheses, quoted arithmetic, decimal-looking versions, variable equations, and code-like equality. Additional engine tests cover division by zero, excessive exponents and values, disallowed calls/names/indexing, and unary-only candidates. Controls are scored for false intervention and component behavior, not arithmetic answer accuracy.

5.3 Model and seed sweep

The primary manifest pins one matched Qwen3 family at 0.6B, 1.7B, and 4B parameters and uses seeds 17, 29, and 41. Repository commit hashes are stored in the configuration and copied into run manifests. The model list is an initial controlled family, not a claim of model-family generality. A second family is required before making architecture-general claims.

Generation is greedy in the initial adapter so seed variability mainly reflects future sampled configurations and experimental plumbing; if sampling is enabled, temperature, top- p , and all random states must be pinned. The correctness-first adapter recomputes the full prefix per generated token after reinjection. This is intentionally inefficient but semantically transparent. Optimized KV-cache interception is outside Paper 1 and must not be silently credited to it.

6 Metrics and Analysis

End task. Report exact rational answer accuracy with 95% confidence intervals. If a model states a final answer after receiving a calculator result, score the final claim, not merely the correct engine value. Report correction, override, and result-use rates separately.

Components. Report intended-expression exposure, candidate recognition, full-expression selection, normalization acceptance/correctness, engine success/correctness, reinjection success, result use, and later override separately. Also report false-intervention rate, intervention count, and state-item count. Oracle differences localize missed or malformed interface events.

Cost. Report prompt, model-generated, and reinjected tokens separately; model calls; engine and end-to-end wall time; and serialized trace/state growth. Do not present the correctness-first Hugging Face adapter’s full-prefix recomputation as a production latency estimate.

Scaling and inference. Report every operator-count by digit-width cell for each model, condition, and seed. Estimate condition-by-difficulty and condition-by-model-size interactions, with paired comparisons because all conditions share examples. Predeclare one primary endpoint and correct or label exploratory multiple comparisons. Preserve raw predictions and event traces so component errors can be reclassified.

7 Frozen Next-Iteration Protocol

The current interface freeze is versioned as `paper1_hard_interfaces_v2_concise`. It retains `strict_arithmetic_v1` unchanged and adds `normalized_arithmetic_v1`, `surface_normalizer_`

`v1`, and `calculator_block_v1`. The normalizer maps allowlisted multiplication, division, Unicode-minus, bracket, and spacing aliases before delegating to the existing `ccpu.arithmetic.postfix.v1` parser. Unknown commands, prose, variables, quoted arithmetic, and quoted or inline fence examples remain inert.

The block grammar is a line-anchored Markdown-like fence named `calculator`. Its contents are collected incrementally, but no candidate is emitted until the closing fence is complete. Thus a valid intermediate line cannot execute prematurely. At close, the same surface normalizer and bounded calculator are used, and a compact typed result is inserted.

The adaptive pilot runs only the LLM-only condition over all 48 cells. Cell selection uses aggregate exact match to choose regimes near 95%, 65%, 30%, and 0%; it never examines assisted outcomes. Once those cells are selected, their definitions, prompts, grammar, normalizer, and diagnostics remain frozen. A distinct dataset seed then generates the held-out comparison across all seven conditions.

Every arithmetic prediction records expression exposure, candidate recognition, full-expression selection, normalization, execution, reinjection, result use, and later override independently. Cost records include prompt, generated, and reinjected tokens, model calls, wall and engine time, serialized trace/state bytes, and deterministic invocation-delimiter characters. Scripted smoke output is marked `empirical: false` and is plumbing evidence only.

7.1 Adaptive calibration status

Two baseline-only XPU calibrations exposed output censoring before any assisted condition was run. Both used Qwen3-0.6B, 96 arithmetic examples over 48 cells, two nominal generation seeds, and a 48-token ceiling, for 192 generations each. Under the original prompt, all 192 generations reached the ceiling and 184 had no scorable final answer. A uniformly applied concise-answer prompt reduced cap hits to 170 and missing answers to 156, but both versions yielded only 3.125% overall exact match per seed. The concise run had 46 cells at 0%, one at 50%, and one at 100%, so it cannot supply the four intended difficulty bands.

These runs are preserved as empirical calibration failures, not treated as the hard-arithmetic comparison. Since greedy argmax decoding produced identical text for every one of the 96 cross-seed pairs, the revised calibration removes the redundant generation seed, broadens the grid downward to one-four operand digits and one-four operators, uses two independently generated examples per cell, and raises the uniform ceiling to 96 tokens.

The revised calibration completed 96 baseline-only generations with 30.2% overall exact match; 78 generations had a scorable final answer. Across 48 cells, 9 reached 100% accuracy, 11 reached 50%, and 28 reached 0%. We froze four whole cells with complete answer coverage and no cap hits: one operator/one digit with a left chain (100%), two operators/two digits with nesting (50%), one operator/three digits with a multiplication tree (50%), and one operator/four digits with a left chain (0%). The two 50% cells are explicitly coarse approximations to the overlapping moderate and transition targets.

The selection record stores source dataset and prediction hashes and states that no assisted outcome was consulted. A separately seeded held-out dataset uses four examples per selected cell, cycling ASCII, LaTeX, Unicode, and bracket surfaces, plus all 12 adversarial controls. The held-out model run remains ungenerated at this protocol-freeze point; the results below were produced only after that dataset, selection record, and interface code were committed.

Table 3: Held-out exact-match accuracy. Intervals are 95% Wilson intervals over 16 arithmetic examples. Paired gains/losses and unadjusted exact McNemar p values compare each condition with LLM-only.

Condition	Accuracy [95% CI]	Gains–losses	Exact p
LLM only	0.438 [0.231, 0.668]	—	—
Matched prompt	0.500 [0.280, 0.720]	1–0	1.000
Explicit tool	0.625 [0.386, 0.815]	3–0	0.250
Strict reflex	0.375 [0.185, 0.614]	0–1	1.000
Normalized reflex	0.750 [0.505, 0.898]	5–0	0.0625
Calculator block	0.125 [0.035, 0.360]	2–7	0.180
Oracle	0.875 [0.640, 0.965]	7–0	0.0156

Table 4: Held-out assisted-interface diagnostics over arithmetic examples. Selection is correctness conditional on an accepted request; use is conditional on successful execution. All conditions had zero false interventions on 12 controls, and engine correctness was 1.0 whenever executed.

Condition	Exposure	Recognition	Selection	Normalize/execute	Use	Override
Explicit tool	0.875	1.000	1.000	0.625	1.000	0.000
Strict reflex	0.625	0.313	0.600	0.313	0.800	0.200
Normalized reflex	0.750	0.750	1.000	0.750	1.000	0.000
Calculator block	0.375	1.000	1.000	0.125	1.000	0.000
Oracle	—	1.000	1.000	1.000	0.875	0.125

8 Held-Out XPU Comparison

8.1 Setup and end-task result

The held-out run used Qwen3-0.6B at revision `c1899de`, PyTorch 2.13.0+XPU, one greedy generation seed (17), and a uniform 96-token ceiling. Four newly generated examples in each of the four frozen cells gave 16 arithmetic examples; 12 controls were crossed with the same seven conditions. All 196 prediction rows report `empirical: true` and `device: xpu`.

Normalized reflex was numerically the strongest non-oracle condition, improving by 31.25 percentage points over LLM-only and 12.5 points over explicit tools. The five paired gains and zero losses are suggestive but do not reach 0.05 in the unadjusted exact test; no multiplicity correction is applied. Strict reflex lost one paired example and gained none. The four held-out cells contain only four examples each and do not show a stable monotone difficulty interaction: normalized-reflex cell accuracies were 1.00, 0.25, 1.00, and 0.75. H1 is therefore unsupported for the original strict watcher, while the normalized variant is a replication candidate rather than a confirmed claim.

8.2 Interface decomposition

Normalized reflex reached and correctly selected the full expression in 12 of 16 cases; all 12 exact results were reinjected, used, and not overridden. Strict reflex executed five candidates, but only three represented the intended full expression. Explicit tool calls were recognized in every arithmetic generation, but six arguments failed the strict parser, primarily because the model copied LaTeX or Unicode notation.

The semantic block result falsifies H3 for the tested zero-shot prompt. All 16 closing fences were recognized, but 14 blocks contained the literal placeholder `EXPRESSION` or mixed instructions/prose

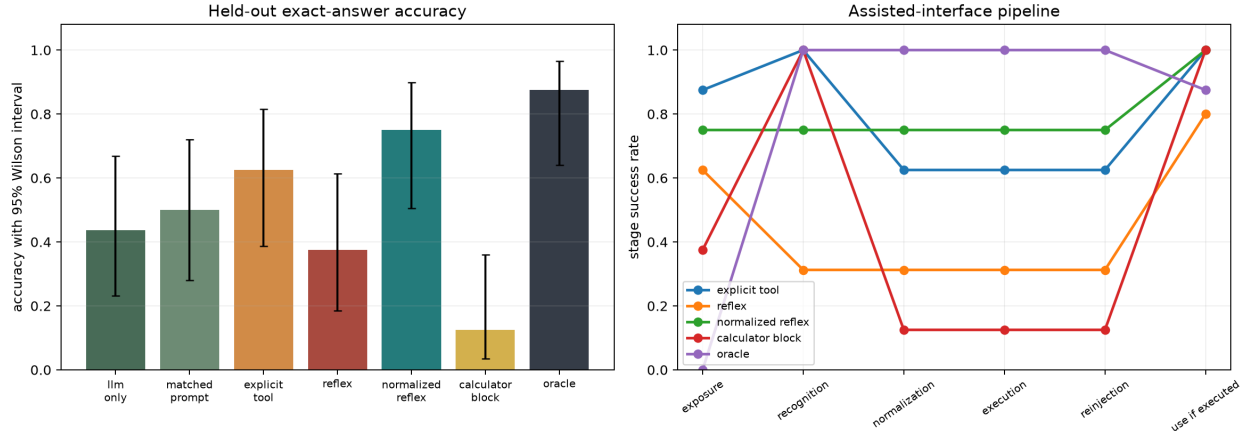


Figure 1: Held-out accuracy (left) and assisted-interface stage decomposition (right). Exposure is model-stream exposure and is not applicable to oracle priming. Use is conditional on execution, which explains high use despite low block normalization.

with arithmetic. Only two blocks normalized and executed; both engine results were exact and used. Thus the block boundary prevented premature execution, but its demonstration-style prompt elicited invalid block contents. This is a generation/normalization failure, not a calculator or closing-fence detection failure.

Cost. Mean generated tokens were 43.1 (LLM-only), 46.1 (matched), 19.2 (explicit), 47.9 (strict), 22.4 (normalized), 32.7 (block), and 15.6 (oracle). Mean wall times followed the correctness-first decoder rather than production serving: 8.0, 8.8, 3.2, 9.6, 4.0, 5.9, and 2.7 seconds, respectively. Normalized reflex used fewer delimiter characters than explicit calls, but it did not beat explicit tools on every measured cost: explicit generated fewer tokens and was faster. H2 is therefore not supported as a quality–cost frontier claim.

8.3 Initial evidence-gate decision

At that point the decision was **no-go for Paper 2**. The original strict reflex does not improve held-out accuracy, the block interface fails zero-shot content elicitation, and the normalized-reflex advantage is unreplicated with $p = 0.0625$ over 16 pairs. A next Paper 1 replication should hold `surface_normalizer_v1` fixed, increase examples and generation seeds under sampling or add a second checkpoint family, and predeclare normalized reflex versus both LLM-only and explicit tool as primary comparisons. Block prompting or lightweight training is now eligible for a separate development split, but must not be tuned on this held-out set.

9 Few-Shot Blocks and Capability Diagnostic

9.1 Endpoint audit and prompt development

The original exact-match labels are preserved. A parallel, condition-independent endpoint extractor recognizes terminal forms such as **Response:** N without using condition identity. Under this audit the original zero-shot block’s answer accuracy changes from 12.5% to 43.8%, while normalized reflex

Table 5: Frozen ICL-G calculator blocks. The 0.6B and 1.7B rows use 16 arithmetic and 12 control examples; the 4B row is a four-arithmetic/two-control smoke only.

Model	Phase	Execute	Semantic	Answer	False controls	Use
Qwen3-0.6B	held-out diagnostic	1.000	1.000	1.000	2/12	1.000
Qwen3-1.7B	held-out diagnostic	0.938	0.938	0.938	0/12	1.000
Qwen3-4B	smoke	1.000	1.000	1.000	0/2	1.000

remains 75.0%. This is an extraction correction only: valid zero-shot block execution remains 12.5%, so the protocol conclusion is unchanged.

We tested six stronger block prompts on a separate developmental split containing four arithmetic examples and four controls. Concrete and verbatim-copy examples made arithmetic execution reliable but initially caused false blocks on all four controls; a positive/negative variant suppressed false blocks but also suppressed arithmetic blocks. A diagnostic with fully seeded fences isolated payload copying. The final order control, ICL-G, puts a negative demonstration first and a concrete positive block last. It achieved 100% semantic payload equivalence, execution, result use, and answer accuracy with no developmental false block. Exact-string payload match was 0% because the model removed redundant parentheses; the bounded parser established semantic equivalence. We froze this prompt as `paper1_calculator_block_icl_v2_order_control` before cross-model runs.

9.2 Matched-family XPU results

Table 5 compares the frozen prompt without per-model tuning. Qwen3-0.6B used the original held-out arithmetic/control set. The 1.7B checkpoint used all seven conditions over the same 28 examples, producing 196 rows. The 4B result is only the predeclared four-arithmetic/two-control smoke gate.

For 0.6B, ICL-G gained nine arithmetic examples and lost none against the preserved LLM-only run ($p = 0.0039$); however, two false blocks fail the safety part of the frozen gate. For 1.7B, it gained nine and lost one ($p = 0.0215$), while normalized reflex gained six and lost one ($p = 0.125$). The sole 1.7B block failure omitted the wrapper and emitted a bare expression. The endpoint audit leaves ICL-G at 93.8%. The unexpectedly low 1.7B oracle score (12.5% originally, 25.0% under endpoint rescoring) reflects model integration after seeded results, not calculator failure; it is retained rather than normalized away.

Family coverage and runtime. Official Llama 3.2 and Gemma checkpoints require manually accepted Hugging Face licenses. No authenticated token was available during this matched-family sweep, so four configured checkpoints were skipped and no community mirror was substituted; this capability-scaling phase is therefore Qwen-only. Approved Gemma access was supplied later for the adapter-placement phase below. Qwen3-1.7B peaked at 3.57 GB XPU memory and its full run took 3074 seconds. Qwen3-4B loaded and completed the smoke at 8.21 GB peak memory in 494 seconds. The block, accuracy/interface, token-latency, and failure plots are generated from one hashed comparison configuration and machine-readable table.

9.3 Current evidence and LoRA gate

The new result reopens the original interface gate: a frozen semantic block is reliable on the 1.7B diagnostic and clean in the 4B smoke. This is sufficient to motivate later heterogeneous-interface

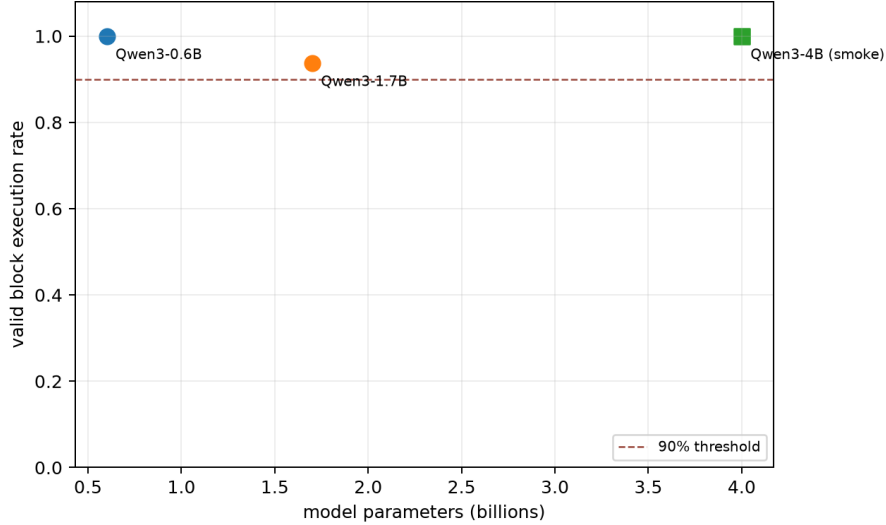


Figure 2: Valid execution under the one frozen ICL prompt. Circles are 16-example held-out diagnostics; the square is a four-example smoke and must not be read as a comparable estimate. Parameter count is descriptive, not a fitted scaling law.

work, while confirmatory Paper 1 claims remain gated on a larger untouched set and additional families. It also places the LoRA question in Regime A: can 0.6B retain its perfect arithmetic protocol while eliminating false control blocks? This asks where interface knowledge should live: transiently in context, persistently in weights, or deterministically in the runtime.

Frozen adapter protocol. The minimal protocol-only experiment uses 80 arithmetic and 80 negative-control requests; a disjoint development set has 20 of each. Arithmetic targets contain only the requested expression inside a calculator fence, never its answer. They cover three–six digit operands, nesting, mixed operations, exact division, long intermediates, and ASCII, LaTeX, Unicode, and bracket surfaces. Controls cover versions, dates, ranges, quoted arithmetic, code, identifiers, and variable-bearing equations. A mechanical audit finds no exact expression or operand overlap among training, development, and the 16-example held-out benchmark.

The XPU training gate completed PEFT forward, backward, and AdamW steps on all three models. The frozen stack adds PEFT 0.20.0 and Accelerate 1.14.0 to PyTorch 2.13.0+XPU and Transformers 4.57.6. LoRA rank is 8 over attention projections, with two epochs and assistant-target-only loss. Qwen3-0.6B is primary. The predeclared ungated second family is the official Apache-2.0 HuggingFaceTB SmolLM2-1.7B-Instruct checkpoint, not a community mirror. A later authenticated run uses the approved official Gemma3-1B checkpoint at pinned revision `dcc83ea`; Llama 3.2 access remains unavailable.

Evaluation compares base+frozen ICL (context), adapter+minimal instruction (weights), and base+normalized reflex (runtime), with base+minimal and adapter+ICL controls. It reports protocol decomposition, false blocks, answer accuracy, prompt/output tokens, training parameters/tokens/time, memory, and inference overhead.

Qwen3-0.6B adapter result. The primary adapter trained in 394.5 seconds on XPU, with 2,293,760 trainable parameters (0.383% of the model), 7,362 assistant target tokens over two epochs, and 1.48 GB peak memory. Mean training loss fell from 0.0526 to 0.000032 and development loss

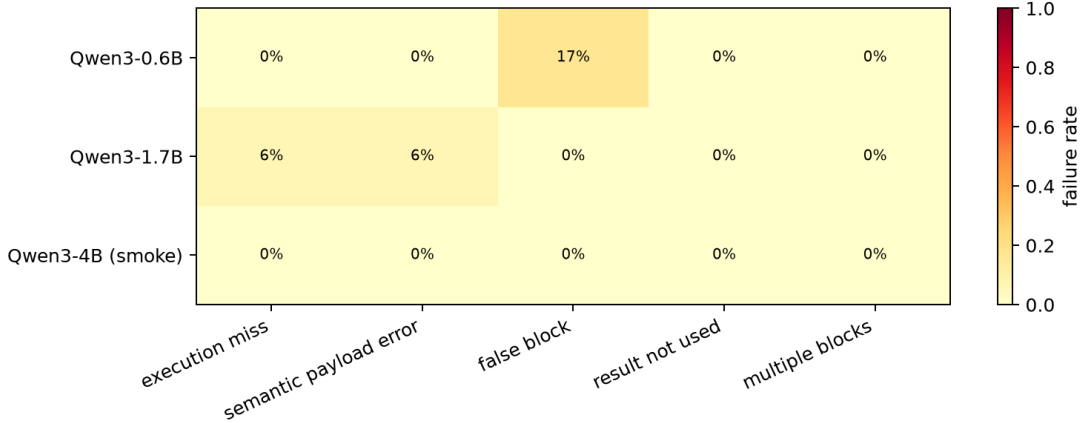


Figure 3: Block failure decomposition. The small model’s remaining error is control discrimination, whereas the 1.7B model has one arithmetic protocol miss.

Table 6: Qwen3-0.6B interface-knowledge placement on 16 arithmetic examples and 12 controls. Execute is valid arithmetic execution; false is the number of control interventions. Tokens are means over all 28 examples.

Placement/condition	Execute	Answer	False	Prompt tok.	Output tok.
Base + minimal	0.000	0.000	0/12	86.6	17.2
Context: base + ICL-G	1.000	1.000	2/12	113.8	18.9
Weights: LoRA + minimal	1.000	1.000	0/12	86.6	14.9
Weights + context: LoRA+ICL	1.000	1.000	4/12	113.8	19.2
Runtime: normalized reflex	0.750	0.750	0/12	61.0	22.4

from 0.000050 to 0.000021. The serialized adapter is 9.2 MB.

Base+minimal emitted no valid arithmetic block, so the instruction alone does not explain the adapter result. LoRA+minimal produced 16/16 semantically correct and executable payloads, used every result, answered all 16 correctly, and opened no block on 12 controls. Payload strings differed harmlessly on 14 cases, primarily by removing redundant parentheses. The adapter adds about 9.2 MB of resident model memory; measured peak inference memory increased from 1.28 to 1.42 GB. Its minimal prompt uses 27.1 fewer tokens than ICL-G and generates four fewer tokens on average.

Context and weights are not additive here. Base+ICL-G had two false blocks; LoRA+ICL-G had four, including quoted arithmetic, quoted fences, pseudocode, and quoted code. Thus the primary Qwen result favors storing the learned semantic selection/serialization contract in adapter weights with a minimal explicit contract, while retaining deterministic execution in the runtime. The normalized runtime remains safer without training but has only 75% expression exposure and execution. No result implies that LoRA learned arithmetic: every answer still comes from the unchanged bounded calculator.

Cross-family replication and placement answer. SmolLM2-1.7B used the identical data and optimization schedule. Its adapter has 3,145,728 trainable parameters (0.183%), is 12.6 MB, used 7,542 target tokens, trained in 768.0 seconds, and peaked at 3.59 GB. Training/development loss changed from 0.1287/0.0049 after epoch one to 0.0013/0.00015 after epoch two.

Table 7: Primary placement comparison. Recurring tokens sum mean prompt and generated tokens. All rates use 16 arithmetic and 12 control examples.

Model	Placement	Interface	Answer	False	Tokens
Qwen3-0.6B	Context	1.000	1.000	0.167	132.7
	Weights	1.000	1.000	0.000	101.5
	Runtime	0.750	0.750	0.000	83.5
SmolLM2-1.7B	Context	0.938	0.875	0.083	184.2
	Weights	1.000	1.000	0.000	127.6
	Runtime	0.938	0.813	0.000	96.6
Gemma3-1B	Context	0.312	0.375	0.083	131.2
	Weights	1.000	0.938	0.000	102.1
	Runtime	0.250	0.312	0.083	86.8

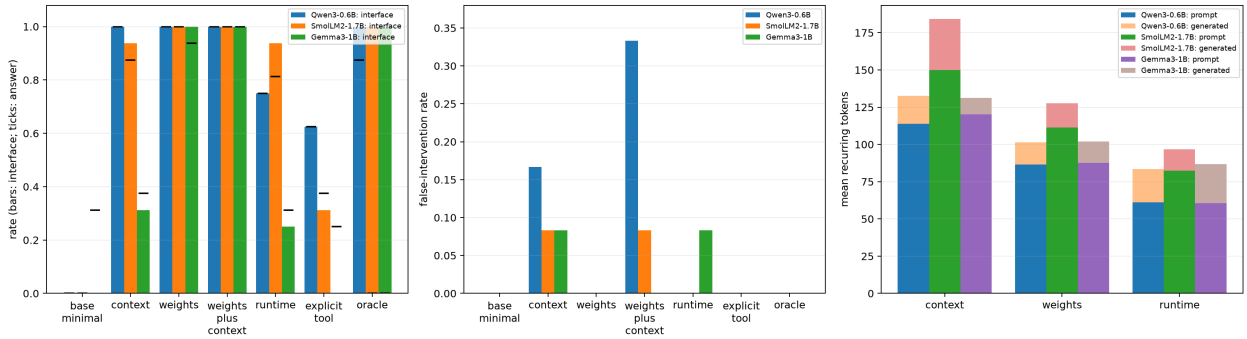


Figure 4: Interface success and answer accuracy (left), false interventions (center), and recurring prompt-plus-output tokens (right). Adapter+minimal has perfect execution and observed control safety in all three families; Gemma needs residual ICL to recover its one copied-expression error.

Base+minimal again emitted 0/16 valid blocks. Base+ICL-G executed 15/16, answered 14/16, and falsely blocked one quoted-fence control. LoRA+minimal executed and answered 16/16, used every exact result, and had 0/12 false blocks. LoRA+ICL-G also answered 16/16 but retained the quoted-fence false block and emitted multiple blocks twice. Normalized runtime executed 15/16 with no false intervention but answered 13/16 because exact result integration was imperfect.

Gemma3-1B used the same data, schedule, and rank, with its native BF16 dtype. Its 1,490,944 trainable parameters (0.149%) serialize to 6.0 MB; training used 7,362 target tokens, took 365.0 seconds, and peaked at 2.52 GB. Training/development loss changed from 0.1654/0.000094 after epoch one to 0.000069/0.000021 after epoch two. Base+minimal emitted 0/16 valid blocks. Base+ICL-G executed 5/16 and falsely blocked one control; normalized runtime executed 4/16 and also had one false intervention. LoRA+minimal executed 16/16 with 0/12 false blocks but copied one expression incorrectly, yielding 15/16 answers. LoRA+ICL-G recovered 16/16 answers and retained 0/12 false blocks. Oracle detection executed all 16 exact calculations, but Gemma used none of the reinjected results, demonstrating that runtime enforcement cannot stop at detection, execution, and reinjection.

The manual autoregressive backend originally stopped only on the tokenizer’s single EOS identifier. Gemma declares both EOS and `<end_of_turn>` in its model generation configuration. Corrected generation honors the full set {1,106}; the corrected base and adapter summaries are the only Gemma sources in this analysis. Earlier v1 diagnostic runs are retained but excluded.

The answer is therefore not one location. Stable semantic selection and block serialization are most reliable in weights under a short explicit contract in all three models. New or changing interfaces can begin in context without training, but demonstrations recur on every request and produced more false interventions; residual context repaired Gemma’s one adapter copying error but was redundant or harmful for Qwen and SmolLM. Parsing, resource bounds, exact arithmetic, reinjection, and enforced result use must remain in the runtime. This division stores policy in weights and mechanism in the runtime, with context serving development, cold start, and measured model-specific residuals. It is a developmental design result, not evidence that all interface knowledge should be fine-tuned.

10 Public GSM8K Transfer Stress Test

10.1 Frozen protocol

GSM8K contains linguistically diverse grade-school word problems with annotated multi-step arithmetic rationales [6]. We selected 120 test examples by the lowest precomputed selection key within three registered strata: 40 with at most two annotated operations, 40 with three or four, and 40 with five or more. The selection artifact redistributes only identifiers, labels, difficulty metadata, and content hashes. Questions are materialized from the pinned source at run time and rejected if their hashes change.

Qwen3-0.6B at revision `c1899de` was evaluated once with greedy seed 23011, FP16 XPU, thinking disabled, and a 160-token ceiling. Seven conditions share the same selected IDs: LLM-only, matched ICL, voluntary generic `__compute`, base calculator block, normalized automatic runtime, oracle calculator ledger, and the retained rank-8 calculator-block LoRA. Generic compute and automatic runtime use the same bounded calculator. The oracle path does not trust source-side arithmetic values: it normalizes and re-executes every supported annotated expression through that calculator before constructing the ledger. Of 425 annotated operations, 382 (89.9%) are supported; 113/120 rows have at least one supported operation and 87/120 have full trace coverage.

The primary endpoint is exact final-answer accuracy. Scorable rate records whether the model emits a parseable endpoint. Assistance rate requires at least one accepted execution, and registered-operation recall is multiset overlap with supported gold operations. The prompts were frozen after a three-item plumbing check. In particular, the generic and automatic instructions retain a concrete 7×8 syntax example; demonstration anchoring observed in the full run was not tuned away. GSM8K has no matched no-compute controls or token-level alignment, so false-activation rate and first-wrong-token prevention are undefined.

10.2 Results

The deployable interfaces do not improve over LLM-only. Voluntary generic compute accepts calls on 114/120 rows, but 168 of 225 calls repeat the demonstration expression `7*8`; its duplicate-call and malformed-row rates are 92.5% and 5.0%, respectively, and registered-operation recall is zero. Automatic runtime accepts assistance on 114/120 rows but recalls only 6.0% of supported operations. It gains two and loses eleven pairs versus LLM-only. Among six rows where voluntary compute emits no valid call, automatic runtime rescues none, so the preregistered Automatic Rescue Rate is 0/6.

The base block prompt exactly ties LLM-only at 13/120 but swaps twelve gains for twelve losses and never opens a valid block. Its row is therefore a verbose elicitation baseline, not calculator evidence. LoRA opens a valid block on 79/120 rows, but 109/120 rows contain malformed block

Table 8: Frozen Qwen3-0.6B GSM8K transfer test (120 examples). Op. recall is over calculator-supported annotated operations. Latency is the mean model wall time; interrupt conditions use correctness-first full-prefix decoding and are not production serving estimates.

Condition	Accuracy	Scorable	Assist	Op. recall	Tokens	ms
LLM only	0.108	1.000	0.000	0.000	18.7	2,776
Matched ICL	0.033	1.000	0.000	0.000	10.7	1,679
Generic compute	0.025	1.000	0.950	0.000	39.8	5,963
Base block prompt	0.108	0.383	0.000	0.000	141.1	42,396
Automatic runtime	0.033	0.992	0.950	0.060	43.9	13,148
Oracle ledger	0.758	0.983	0.942	1.000	39.9	5,907
LoRA block	0.008	0.075	0.658	0.065	55.9	20,457

activity, operation recall is 6.5%, and only nine rows emit a scorable endpoint. It answers one row correctly. The same adapter’s 16/16 synthetic held-out result therefore learned a narrow selection/serialization distribution rather than GSM8K decomposition.

The oracle ledger reaches 91/120 (75.8%, 95% Wilson interval 67.4–82.6%), with 79 paired gains and one loss versus LLM-only. This establishes large arithmetic-and-integration headroom when formalization is supplied, not a deployable routing result. Accuracy is not monotone in operation-count strata: oracle scores 60.0%, 87.5%, and 80.0%, so annotated operation count alone is not semantic difficulty.

This stress test changes the placement conclusion. Context or small adapter weights can store a stable local syntax contract, and the runtime can guarantee bounded exact execution, but none supplies the missing semantic decomposition on public word problems. Paper 1 therefore demonstrates synthetic interface feasibility and oracle public headroom, not calculator-assistance transfer.

11 Preliminary XPU Pilot

11.1 Setup

We ran a diagnostic pilot with Qwen3-0.6B at pinned revision `c1899de` (the full hash is stored in the manifest), PyTorch 2.13.0+XPU, and an Intel integrated GPU. The deterministic sample crossed one and three operators with one- and two-digit operands, using addition and multiplication. It contained two examples per cell (eight arithmetic examples) and eight matched non-trigger controls. One generation seed (17) and five conditions yielded 80 generations. The 96-token output budget was selected after a shorter plumbing run exposed truncation; the shorter run is not included in the results. Confidence intervals are Wilson intervals over only eight arithmetic observations per condition and are therefore intentionally wide.

11.2 Results

The reflex condition did not improve exact-match accuracy over LLM only in this pilot. Explicit-tool accuracy was numerically higher and oracle intervention was perfect, but the sample is too small for either difference to support a general claim. No condition falsely intervened on a control. Accepted explicit-tool candidates represented the intended expression in every triggered case, whereas only 50% of reflex candidates did. Inspection showed that the strict watcher often intercepted a valid intermediate subexpression rather than the full task expression. Once a candidate reached the engine, execution correctness was 100% in both conditions. Reflex result-use was 66.7%,

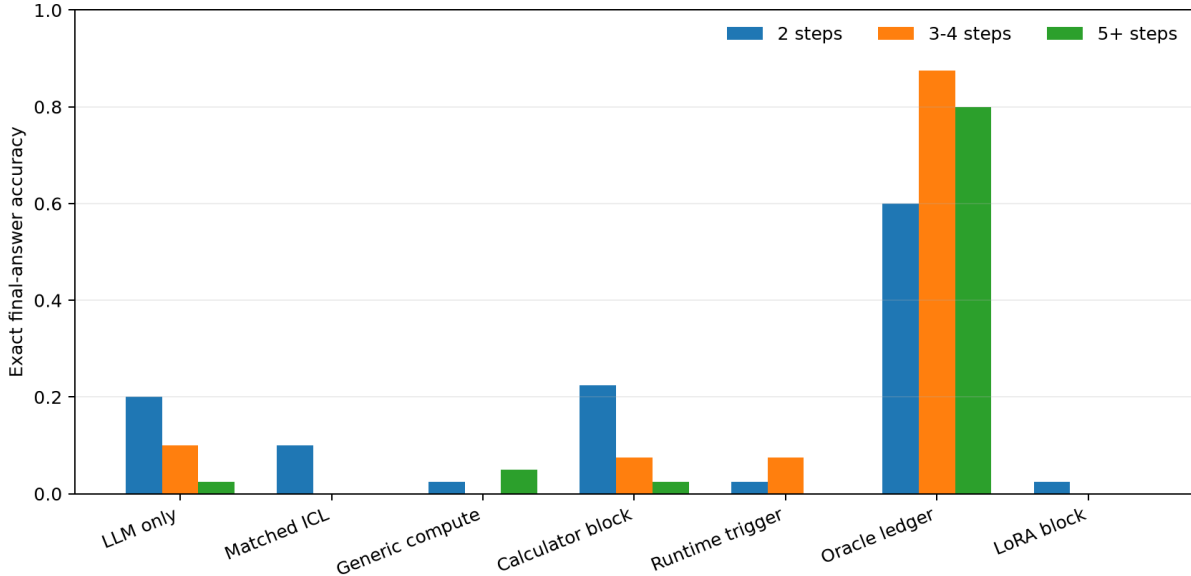


Figure 5: Exact GSM8K accuracy by registered operation-count stratum. Each bar contains 40 examples. Only the oracle ledger transfers strongly; deployable interfaces remain near zero across strata.

Table 9: Exploratory XPU pilot. Accuracy intervals are 95% Wilson intervals; tokens and latency are means. Latency reflects the correctness-first full-prefix adapter and is not a production estimate.

Condition	Accuracy	Trigger recall	Output tokens	Latency (ms)
LLM only	0.750 [0.409, 0.929]	—	40.56	8028
Matched prompt	0.625 [0.306, 0.863]	—	45.81	9380
Explicit tool	0.875 [0.529, 0.978]	0.625	22.50	4410
Reflex	0.750 [0.409, 0.929]	0.750	46.06	9195
Oracle	1.000 [0.676, 1.000]	1.000	15.31	2718

and one third of used results were later overridden. These component results locate the present bottleneck in exposure, selection, and downstream use rather than arithmetic execution.

At this protocol revision H1 is not supported: the automatic reflex tied the unassisted baseline and failed to dominate explicit invocation. H2–H4 remain untested because this pilot includes neither a production interface nor the required model-size and seed sweeps. Paper 2 therefore remains gated. A confirmatory run must increase examples per cell, use all declared seeds and model sizes, add a second family, and freeze any detector change before examining outcomes.

12 Falsification and Evidence Gate

The reflex claim is not supported if any of the following holds under matched capability and declared cost accounting:

- answer gains disappear when final model claims, rather than inserted engine values, are scored;
- false triggers, normalization errors, or ignored results offset exact execution gains;

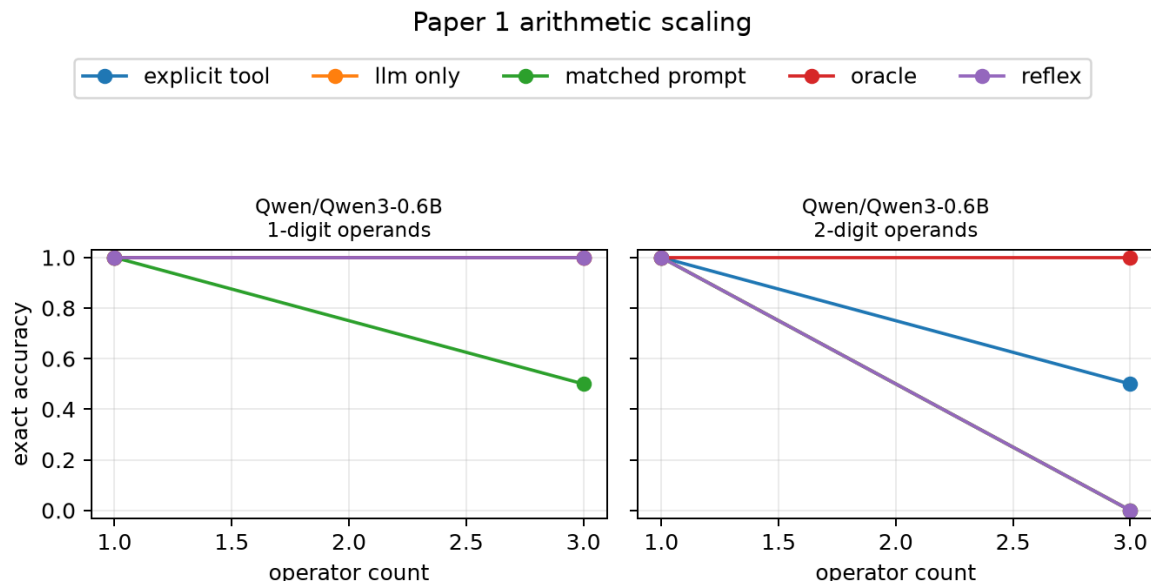


Figure 6: Exploratory exact-match accuracy by generated difficulty cell. Each point contains only two examples; the figure diagnoses plumbing and candidate behavior and must not be read as a scaling estimate.

- explicit calculator calling matches or dominates the quality–token–latency frontier;
- reflex and baseline accuracy have indistinguishable difficulty slopes;
- gains occur only with oracle detection or only in one easy/model cell; or
- results fail to reproduce across seeds or a second model family.

Paper 2 is justified only if Paper 1 identifies a reproducible regime where automatic strict assistance helps and mixed operation types expose a limitation of one calculator. Paper 3 is not justified by syntax brittleness alone; learned interrupts require a measured task regime worth extending.

13 Reproduction

The reusable runtime lives under `src/ccpu/common`; Paper 1 code lives under `src/ccpu/paper1`. From the repository root, the dependency-free protocol check is:

```
python -m pip install -e .
python -m pytest
python -m ccpu paper1 generate \
    --config configs/paper1/smoke.json \
    --output artifacts/paper1/smoke/dataset.jsonl
python -m ccpu paper1 validate \
    --dataset artifacts/paper1/smoke/dataset.jsonl
python -m ccpu paper1 simulate \
    --dataset artifacts/paper1/smoke/dataset.jsonl \
    --output-dir artifacts/paper1/smoke/run
```


The pinned pretrained path is:

```
python -m pip install -e ".[hf]"
python -m ccpu paper1 generate \
  --config configs/paper1/primary.json \
  --output artifacts/paper1/primary/dataset.jsonl
python -m ccpu paper1 run-hf \
  --dataset artifacts/paper1/primary/dataset.jsonl \
  --config configs/paper1/primary.json \
  --output-dir artifacts/paper1/primary/run
```

The reported XPU pilot uses the direct interpreter from the validated XPU environment and can be reproduced with:

```
$py = 'C:\Users\j.simao\.venvs\modal-llm-xpu\Scripts\python.exe'
$env:PYTHONPATH = (Resolve-Path src).Path
& $py -m ccpu paper1 generate \
  --config configs/paper1/xpu_pilot.json \
  --output artifacts/paper1/xpu_pilot/dataset.jsonl
& $py -m ccpu paper1 run-hf \
  --dataset artifacts/paper1/xpu_pilot/dataset.jsonl \
  --config configs/paper1/xpu_pilot.json \
  --output-dir artifacts/paper1/xpu_pilot/run96
& $py -m ccpu paper1 replay \
  --dataset artifacts/paper1/xpu_pilot/dataset.jsonl \
  --completions artifacts/paper1/xpu_pilot/run96/predictions.jsonl \
  --output-dir artifacts/paper1/xpu_pilot/final
```

The held-out comparison and derived analysis use:

```
& $py -m ccpu paper1 run-hf \
  --dataset artifacts/paper1/hard_heldout_xpu/dataset.jsonl \
  --config configs/paper1/hard_heldout_xpu.json \
  --output-dir artifacts/paper1/hard_heldout_xpu/run \
  --device xpu
& $py -m ccpu paper1 paired \
  --dataset artifacts/paper1/hard_heldout_xpu/dataset.jsonl \
  --predictions artifacts/paper1/hard_heldout_xpu/run/predictions.jsonl \
  --output artifacts/paper1/hard_heldout_xpu/run/paired_analysis.json
python -m ccpu paper1 plot-interfaces \
  --summary artifacts/paper1/hard_heldout_xpu/run/summary.json \
  --output docs/papers/paper1/paper1_heldout_interfaces.png
```

Device selection for `device: auto` is CUDA, then XPU, then CPU. The replay step applies the current frozen evaluator to preserved generations and records both the source prediction hash and rescore provenance.

Use `--limit`, `--model`, and repeated `--condition` arguments for resource preflight. Each run writes predictions, full traces, summary tables, hashes, configuration, model metadata, environment information, and git revision/dirty status. The replay command permits model generation in a

separate harness while preserving identical controller and evaluation semantics. The `plot` command renders the machine-readable scaling cells as accuracy curves by model, operand width, condition, and operator count. The `compare-models` command consumes the frozen comparison configuration and regenerates the cross-model table plus block-capability, interface-accuracy, token-latency, and failure-mode figures without manually entered results. The `generate-lora-data` and `train-lora` commands implement the versioned protocol-only adapter phase; the former must pass its leakage audit before the latter can run. The `analyze-placement` command hashes the source summaries and training reports and regenerates Table 7 and Figure 4 inputs.

The public transfer run is resumable by condition and is reproduced with:

```
& $py -m ccpu paper1 freeze-public-gsm8k \
  --source-selection artifacts/paper2/public_benchmarks/data_v1/selection.jsonl \
  --output-dir artifacts/paper1/public_gsm8k_v1/data
foreach ($condition in @('llm_only','matched_icl','generic_compute', \
  'calculator_block','runtime_trigger','oracle_calculator', \
  'lora_calculator_block')) {
  & $py -m ccpu paper1 run-public-gsm8k \
    --public-config configs/common/public_benchmarks.json \
    --cache-root C:\Users\j.simao\.cache\ccpu-public-v1 \
    --selection artifacts/paper1/public_gsm8k_v1/data/selection.jsonl \
    --model-config configs/paper1/public_gsm8k_qwen_xpu.json \
    --condition $condition --checkpoint-every 10 \
    --output-dir artifacts/paper1/public_gsm8k_v1/qwen3_0_6b/$condition
}
$analysis = @('paper1','analyze-public-gsm8k','--output-dir', \
  'artifacts/paper1/public_gsm8k_v1/analysis')
foreach ($condition in @('llm_only','matched_icl','generic_compute', \
  'calculator_block','runtime_trigger','oracle_calculator', \
  'lora_calculator_block')) {
  $analysis += @('--predictions', \
    "artifacts/paper1/public_gsm8k_v1/qwen3_0_6b/$condition/predictions.jsonl")
}
& $py -m ccpu @analysis
```

14 Limitations

The elicitation instruction asks reflex models to expose an expression followed by an equals sign. This is less explicit than an API call but still changes generation behavior, and a model that answers directly may never trigger. Both trigger rate and conditional-on-trigger accuracy are therefore necessary. The strict lexical grammar does not understand intent and will miss prose arithmetic; expanding semantics belongs to Paper 3. Quote suppression is deliberately limited, and code-like or unusual formatting remains an adversarial surface.

The synthetic task controls arithmetic complexity but not realistic discourse. The textual explicit-tool baseline may understate a model’s native structured function-calling ability. Full-prefix token recomputation overstates runtime cost. The main comparisons have one greedy seed, 16 arithmetic examples, and four observations per selected cell. Their intervals remain wide, the adaptive bands are coarse, and no architecture or universal scaling generalization is supported. The 4B row is only a six-item smoke. Llama remains unavailable behind manual authorization;

approved Gemma3-1B and ungated SmolLM2 provide two cross-family adapter replications but cannot establish broad family effects. The adapter data are synthetic and small, and near-zero development loss may reflect the narrow protocol distribution. Twelve controls do not tightly bound rare false triggers, as the ICL failures illustrate. Adapter+minimal was selected by the predeclared gate, but the three-family comparison remains developmental rather than confirmatory. The zero-shot block prompt’s historical result is not reinterpreted by the separately developed ICL prompt, endpoint audit, or LoRA result. Finally, exact calculator output cannot repair an incorrectly copied expression, a model that ignores the result, or a task whose premises are wrong. The public transfer slice adds further limits: it has one small checkpoint, one greedy seed, and no no-compute controls. Operation-count strata are not a complete difficulty model. Source rationales provide noisy opportunity annotations, and 10.1% of annotated operations fall outside the registered calculator surface. The concrete syntax example causes strong anchoring in generic and automatic conditions; because it was frozen before the full outcome, this is valid evidence for that protocol but not evidence that all tool or trigger prompts must fail.

15 Related Work

Toolformer trains language models to decide when and how to invoke APIs, including a calculator [1]. PAL delegates model-generated programs to an interpreter [2], while ReAct interleaves model-authored reasoning and actions [3]. These systems establish the value of external execution. Paper 1 isolates a different control point: a runtime recognizes ordinary strict arithmetic already emerging in the output and invokes a bounded engine without a function-name decision. The comparison to explicit invocation is empirical; the paper does not assume that the reflex interface is better.

LoRA freezes base weights and injects trainable low-rank matrices [4]; our use of it is conventional, and the empirical question is only whether a small adapter can store this interface contract. Self-RAG-like systems show that models can learn control or reflection tokens for inference actions [5]. Calculator fences are a typed local execution region, not a claim to have invented trained action syntax or control tokens.

16 Conclusion

Paper 1 reduces the cognitive-coprocessor proposal to its smallest credible experiment. A character-incremental detector, typed allowlisted micro-IR, exact bounded calculator, append-only state, and immediate text reinjection are enough to test whether automatic computation can improve an autoregressive stream. The original strict watcher does not improve accuracy, and the zero-shot fenced block usually emits invalid content. Deterministic surface normalization is numerically stronger, executing 12 of 16 full expressions and improving five paired examples without a loss, but remains an unreplicated result. Separate development shows that the block failure was not intrinsic to the grammar: one frozen two-shot prompt reaches 100% arithmetic execution on Qwen3-0.6B and 93.8% on Qwen3-1.7B. Capability changes the error type: 0.6B falsely blocks two controls, while 1.7B has no false block and one arithmetic miss. A clean 4B smoke supports feasibility but not scaling. Paper 1 therefore establishes a plausible few-shot semantic execution interface and a protocol-only adapter result across three small-model families. Adapter+minimal reaches perfect execution and observed control safety in all three; it reaches perfect answers in Qwen and SmolLM and 15/16 in Gemma, where residual ICL restores 16/16. Base+minimal never opens a valid arithmetic block in any family. Context enables cold-start behavior but costs more tokens and can produce false interventions; after adaptation it is redundant or harmful except for the measured Gemma resid-

ual. The normalized runtime is training-free but remains limited by model expression exposure and result integration. Gemma and SmolLM oracle runs further show that exact reinjection alone does not guarantee result use. The resulting architecture is hybrid: context for rapid development and measured residuals, weights for a stable semantic contract, and the runtime for typed parsing, bounds, exact computation, and enforced use. Powered confirmatory data and an authorized Llama run remain necessary before stronger placement or family claims. On public GSM8K, however, this narrow adapter and the training-free automatic interface do not transfer: high activation coexists with 6.5% or lower operation recall and at most 3.3% accuracy, while the bounded-calculator oracle reaches 75.8%. Where interface knowledge should live is therefore conditional: syntax can live in context or weights and exactness in the runtime, but natural-language decomposition is not solved by any of the three in this experiment.

References

- [1] T. Schick, J. Dwivedi-Yu, R. Dessi, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, and T. Scialom. Toolformer: Language models can teach themselves to use tools. *NeurIPS*, 2023. <https://arxiv.org/abs/2302.04761>.
- [2] L. Gao, A. Madaan, S. Zhou, U. Alon, P. Liu, Y. Yang, J. Callan, and G. Neubig. PAL: Program-aided language models. *ICML*, 2023. <https://arxiv.org/abs/2211.10435>.
- [3] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. *ICLR*, 2023. <https://arxiv.org/abs/2210.03629>.
- [4] E. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-rank adaptation of large language models. *ICLR*, 2022. <https://arxiv.org/abs/2106.09685>.
- [5] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. *ICLR*, 2024. <https://arxiv.org/abs/2310.11511>.
- [6] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman. Training verifiers to solve math word problems. *arXiv:2110.14168*, 2021. <https://arxiv.org/abs/2110.14168>.